GenRewrite: Query Rewriting via Large Language Models

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Abstract

Query rewriting is an effective technique for refining poorly written queries before they reach the query optimizer. However, manual rewriting is not scalable, as it is prone to errors and requires deep expertise. Traditional query rewriting algorithms fall short too: rule-based approaches fail to generalize to new query patterns, while synthesis-based methods struggle with complex queries. Fortunately, Large Language Models (LLMs) already possess broad knowledge and advanced reasoning capabilities, making them a promising solution for tackling these longstanding challenges.

In this paper, we present GenRewrite, the first holistic system that leverages LLMs for query rewriting beyond traditional rules. We introduce the notion of Natural Language Rewrite Rules (NLR2s), which serve as hints for the LLM while also a means of knowledge transfer from rewriting one query to another, allowing GenRewrite to become smarter and more effective over time. We present a novel counterexample-guided technique that iteratively corrects the syntactic and semantic errors in the rewritten query, significantly reducing the LLM costs and the manual effort required for verification. Across the standard TPC-DS [8] and JOB [25] benchmarks and their SQLStorm-generated variants [38], GenRewrite consistently optimizes more queries at every speedup threshold than all baselines. At the ≥2x threshold on TPC-DS, GenRewrite improves 25 queries— 1.35x more than LLM-driven baselines and 2.6x more than LLM-enhanced rule-based baselines—and the gap widens further on TPC-DS (SQLStorm); on JOB and its SQLStorm variant, where queries are simpler, absolute gains are smaller but GenRewrite still leads by a notable margin.

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1 Introduction

Inefficient queries are a significant problem for nearly any organization relying on a database system. These poorly written queries—whether authored by inexperienced users or auto-generated by business intelligence (BI) tools—are a major cause of slow performance and, in cloud databases like Snowflake [13] or BigQuery [18], can lead to excessive costs. As a result, SQL-to-SQL query rewriting is a crucial step in optimizing performance and reducing expenses, as the likelihood of the query optimizer finding an efficient query plan is significantly diminished if it is given a poorly-written query to begin with. However, whether performed manually or automatically, query rewriting has remained a challenging task.

Challenges of manual query rewriting. The poorly-written queries that have the biggest impact on the overall cost and performance tend to be quite complex too. For example, it is not uncommon for queries auto generated by Looker [3] to span 100s of lines. Ensuring the rewritten query preserves the semantics of the original requires a deep understanding of the database schema, constraints, and the user intent, and is therefore a time-consuming and error-prone task, even for seasoned database experts. The fact that there are often thousands, if not millions, of such slow queries makes the manual approach a monumental undertaking.

Limitations of traditional automated query rewriting. Despite extensive research, automated query rewriting techniques have long faced fundamental limitations that hinder their practical effectiveness. The most common approach relies on *rewrite rules*—–pattern-matching transformations that replace query fragments with optimized equivalents while preserving semantics. Regardless of whether the rules are developed and provided by experts [10, 12, 35] or are inferred automatically [16, 46], a rule-based query rewriter is as effective as the set of rules provided to it. By definition, rule-based query rewriters[1] are fundamentally incapable of optimizing new query patterns that do not match their existing rules [17]. Although synthesis-based query-rewriting does not require prior rules [17], in practice it only handles simple queries—for instance, SlabCity [17] optimizes just 2 out of 22 TPC-H [9] queries and 3 out of 99 TPC-DS [8] queries.

Large language models (LLMs). The shining success of large language models (LLMs) in performing complex and open-ended tasks has created new hopes for solving some of the hardest database problems as well. While researchers have used LLMs for Text-to-SQL [19, 20, 23, 36, 42] and a few other areas, e.g., data cleaning and integration [33, 41], table processing [28, 30] and database diagnosis [50], their potential for query rewriting has remained largely unexplored. In instances where human experts or existing rules cannot rewrite a query, say for complex queries or new query patterns, could LLMs be leveraged to uncover rewrite opportunities due to their broad general knowledge and advanced reasoning capabilities? If so, this could drastically lower the burden on human experts and allow for a much larger set of queries to benefit from rewriting. To the best of our knowledge, this paper is the first[2] to

[1] Rule-based query *rewriting* should not be confused with rule-based query *optimization*; the latter is simpler, and thus much more common in practice [21].
[2] See §7 for related work published after the initial publication of our work on ArXiv.

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